

# AI and finance

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## Bayesian inference

## Naive example: Bernulli

Suppose to be interested in knowing with which probability the price of a stock will go up or down knowing its value now. Consider a 15-minutes spaced time grid and suppose that the historical movements of the last 12 hours and 30 minutes are given by the following picture.

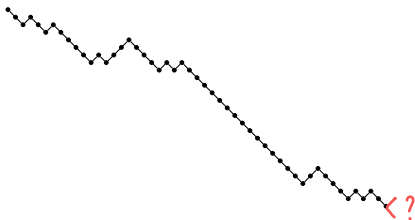


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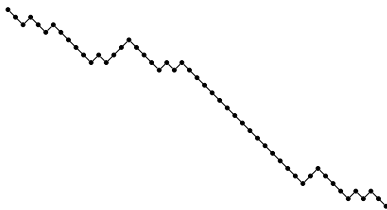


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Observe that this corresponds to 12 upward movements and 38 downward movements. We suppose that each movement is independent from the previous ones but they all share the same distribution.

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$$\mathbb{P}(\text{"up"}) = \mathbb{P}(Y=1) = \theta = \theta^1 (1-\theta)^{1-1}$$

$$\mathbb{P}(\text{"down"}) = \mathbb{P}(Y=0) = 1 - \theta = \theta^0 (1-\theta)^{1-0}$$

$$\Rightarrow \mathbb{P}(Y=y) = \theta^y (1-\theta)^{1-y}$$

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probability that  $Y=y$  knowing  $\theta$ .

  
probability density function



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$$\begin{aligned} p(\mathbf{y}|\theta) &= p(y_1, \dots, y_n|\theta) \stackrel{\text{ind!}}{=} p(y_1|\theta) \cdot \dots \cdot p(y_n|\theta) \\ &= \theta^{y_1} (1-\theta)^{1-y_1} \cdot \dots \cdot \theta^{y_n} (1-\theta)^{1-y_n} \\ &= \prod_{i=1}^n \theta^{y_i} \prod_{i=1}^n (1-\theta)^{1-y_i} \\ &= \theta^{\sum_{i=1}^n y_i} (1-\theta)^{n - \sum_{i=1}^n y_i} \end{aligned}$$

e.g. 1, 1, 0, 0, 1, 0  
up up down down ...

$$\Theta^{\gamma_1} \Theta^{\gamma_2} = \Theta^{\gamma_1 + \gamma_2}$$

$$\begin{aligned} (1-\Theta)^{1-\gamma_1} (1-\Theta)^{1-\gamma_2} &= (1-\Theta)^{1-\gamma_1 + 1 - \gamma_2} \\ &= (1-\Theta)^{2-(\gamma_1 + \gamma_2)} \end{aligned}$$

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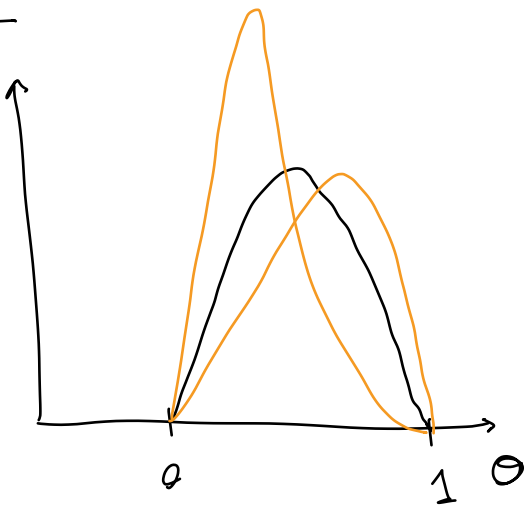
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$\hookrightarrow \theta \sim \mathcal{U}(0,1)?$  ~~Wrong support!~~  
 $\theta$  is a probability  $\in [0,1]$   
 $\hookrightarrow \theta \sim \text{Beta distribution}$

Beta

pdf



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Goal: deduce the shape of the **posterior**  $p(\theta|\mathbf{y})$ , ← probability that the parameter is  $\odot$  given the observations.

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**How?**

## Bayes's rule and Law of total probability

### Lemma (Bayes)

$$p(\theta|y) = \frac{p(y|\theta)p(\theta)}{p(y)}.$$

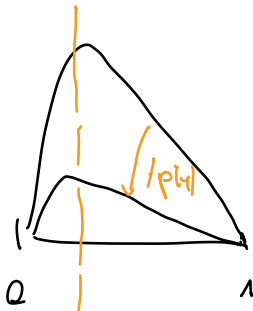
likelihood

prior

not so important!

It does not depend on  $\theta$ .

$p(y)$  is a scaling function that does not impact the shape of the distribution



## Bayes's rule and Law of total probability

Lemma (Bayes)

$$p(\theta|\mathbf{y}) = \frac{p(\mathbf{y}|\theta)p(\theta)}{p(\mathbf{y})}.$$

Lemma (Law of total probability)

$$p(\mathbf{y}) = \int p(\mathbf{y}|\theta)p(\theta)d\theta$$

likelihood

prior



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proportional to

Then, we deduce the shape of the **posterior**

$$p(\theta|\mathbf{y}) \propto p(\mathbf{y}|\theta)p(\theta) \propto \begin{cases} \theta^{\sum_{i=1}^n y_i} (1 - \theta)^{n - \sum_{i=1}^n y_i} \theta(1 - \theta) & \text{if } \theta \in [0, 1], \\ 0 & \text{else.} \end{cases}$$



Example: with numbers

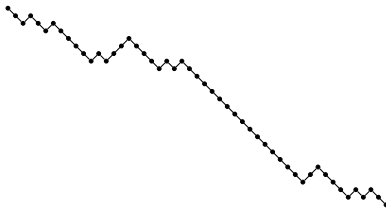


Figure: Price movements



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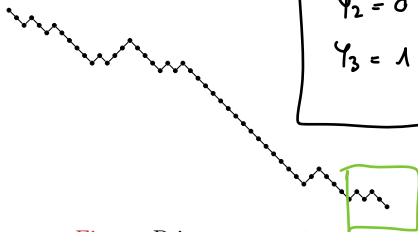


Figure: Price movements

$$\begin{array}{ll}
 y_1 = 1 & y_4 = 0 \\
 y_2 = 0 & y_5 = 0 \\
 y_3 = 1 & 
 \end{array}$$

The last 5 observations reads:  $u, d, u, d, d$ . This translates to

$$n = 5 \quad \text{and} \quad \mathbf{y} = (1, 0, 1, 0, 0)$$

up down up down down

which yields

$$\begin{aligned}
 p(\theta|\mathbf{y}) &\propto p(\mathbf{y}|\theta)p(\theta) \propto \theta^{\sum_{i=1}^5 y_i + 1} (1 - \theta)^{5 - \sum_{i=1}^5 y_i + 1} = \\
 &= \theta^{1+0+1+0+0+1} (1-\theta)^{5-(1+0+1+0+0)+1} = \\
 &= \theta^3 (1-\theta)^4
 \end{aligned}$$

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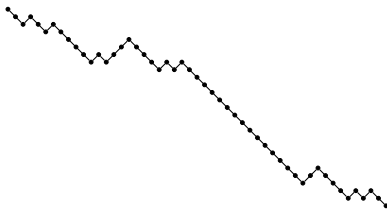


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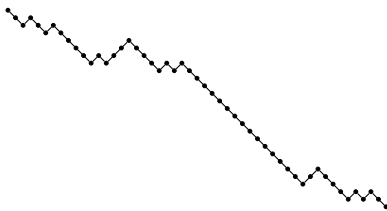


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$$\alpha = 4 \quad \beta = 5$$

↓                  ↓

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The corresponding maximum is attained by  $\hat{\theta} = 3/7 = 0.43$ .

↳ mode of a beta (4,5) distribution



## Example: with numbers

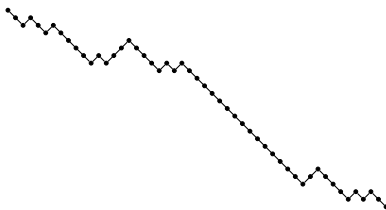


Figure: Price movements

The last 5 observations reads:  $u, d, u, d, d$ . This translates to

$$n = 5 \quad \text{and} \quad \mathbf{y} = (1, 0, 1, 0, 0)$$

which yields

$$p(\theta|\mathbf{y}) \propto p(\mathbf{y}|\theta)p(\theta) \propto \theta^{\sum_{i=1}^5 y_i + 1} (1 - \theta)^{5 - \sum_{i=1}^5 y_i + 1} = \theta^3 (1 - \theta)^4.$$

The corresponding maximum is attained by  $\hat{\theta} = 3/7 = 0.43$ .

Conclusion: we expect the next movement to be up with a prob. of 43%.

Example: with numbers

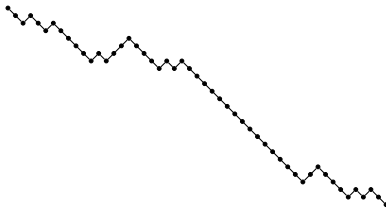


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## Example: with numbers

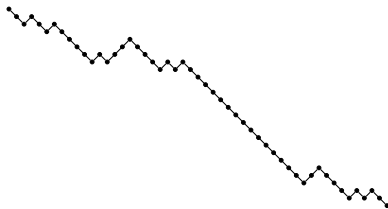


Figure: Price movements

With all the infos (12 ups, 38 downs hence  $\mathbf{y} = (0, 0, 1, \dots, 0, 0)$ ) we get

$$p(\theta|\mathbf{y}) \propto p(\mathbf{y}|\theta)p(\theta) \propto$$



$$\theta^{1 \cdot \theta}$$

$$\theta^{\sum y_i} (1-\theta)^{n - \sum y_i} = \theta^{12} (1-\theta)^{50-12}$$

## Example: with numbers

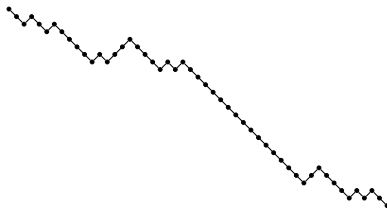


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$$p(\theta|\mathbf{y}) \propto p(\mathbf{y}|\theta)p(\theta) \propto \theta^{13}(1-\theta)^{39}.$$

↑

$$\alpha = 14, \quad \beta = 40$$



## Example: with numbers

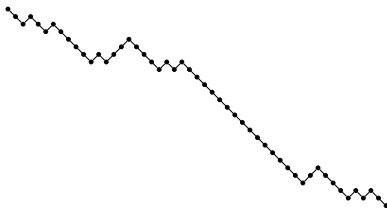


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The corresponding maximum is attained by  $\hat{\theta} = \frac{13}{52} = 0.25$ .

↑  
mode of beta (14, 40)

## Example: with numbers

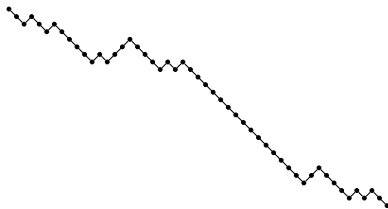


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Conclusion: we expect the next movement to be up with a prob. of 25%.

## Example: with numbers

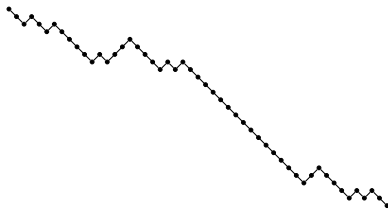


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The corresponding maximum is attained by  $\hat{\theta} = \frac{13}{52} = 0.25$ .

Conclusion: we expect the next movement to be up with a prob. of 25%.

Observation: we computed the posterior distribution which is much more than the **point estimation** given by  $\hat{\theta} = 25\%$ . We computed a **whole distribution** of our believes considering both our prior and the given observation.

## Bayesian estimation

Consider a probabilistic model for a vector of random variables  $\mathbf{Y}$  parametrized with a vector of parameters  $\boldsymbol{\theta}$ .

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Bayesian approach: include in the estimator our a-priori believes on  $\boldsymbol{\theta}$ . Such distribution is called **prior**.

## Bayesian inference from data

Let  $\theta$  be a vector of parameters of some model and let  $\mathbf{y} = y_1, \dots, y_n$  be  $n$  i.i.d. observations of some random variable  $Y$ .

 independent identically distributed random variables.

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- The **prior**  $p(\boldsymbol{\theta})$ .
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  - The **posterior**  $p(\theta|\mathbf{y})$ .

Bayes rule!

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Bayes' rule converts the prior into the posterior incorporating the evidence provided by the observed data.

$$p(\theta|\mathbf{y}) = \frac{p(\mathbf{y}|\theta)p(\theta)}{p(\mathbf{y})}.$$

## Bayesian inference from data

our beliefs before seeing the data

Let  $\theta$  be a vector of parameters of some model and let  $\mathbf{y} = y_1, \dots, y_n$  be  $n$  i.i.d. observations of some random variable  $Y$ .

We consider the following quantities:

- The **prior**  $p(\theta)$ . *independence*
- The **likelihood**  $p(\mathbf{y}|\theta)$ . Since  $y_1, \dots, y_n$  are independent we have  $p(\mathbf{y}|\theta) = \prod_{i=1}^n p(y_i, \theta)$ . *law of total probability*
- The **marginal** likelihood  $p(\mathbf{y}) = \int p(\mathbf{y}|\theta)p(\theta)d\theta$ . Observe that the marginal likelihood does not depend on  $\theta$ .
- The **posterior**  $p(\theta|\mathbf{y})$ . *Our goal: distribution of  $\theta$  given the observation and the prior*

Bayes' rule converts the prior into the posterior incorporating the evidence provided by the observed data.

$$p(\theta|\mathbf{y}) = \frac{p(\mathbf{y}|\theta)p(\theta)}{p(\mathbf{y})}$$

*compute!* *choose*

Since the marginal does not depend on  $\theta$  we also have:

$$p(\theta|\mathbf{y}) \propto p(\mathbf{y}|\theta)p(\theta),$$

where  $\propto$  denotes the proportionality.

## Frequentist inference

## Maximum likelihood estimation

In one sentence: not imposing any prior on the unknown parameter  $\theta$  and studying the likelihood

$$p(\mathbf{y}|\theta)$$

instead of the posterior

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# Maximum likelihood estimation

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instead of the posterior

$$p(\mathbf{y}|\theta)$$
$$p(\mathbf{y}|\theta)p(\theta).$$

To be maximized wrt  $\theta$ !

If simpler, replacing the likelihood with the log likelihood

$$\log(p(\mathbf{y}|\theta)).$$

- $\log(\prod p(y_i|\theta)) = \sum \log(p(y_i|\theta))$
- $\operatorname{argmax}_{\theta} p(y|\theta) = \operatorname{argmax}_{\theta} \log(p(y|\theta))$
- to get rid of the exp.

Example: with numbers

$$P(Y=1|\theta) = \theta$$

$$P(Y=0|\theta) = 1-\theta$$

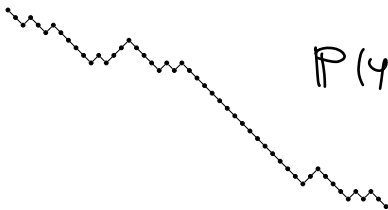


Figure: Price movements

$$P(Y|\theta) = \prod_{i=1}^{50} p(y_i|\theta) = \theta^{*ones} (1-\theta)^{*zeros}$$



## Example: with numbers

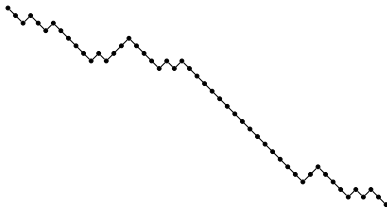


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With all the infos (12 ups, 38 downs hence  $\mathbf{y} = (0, 0, 1, \dots, 0, 0)$ ) we get

$$p(\mathbf{y}|\theta) \propto$$

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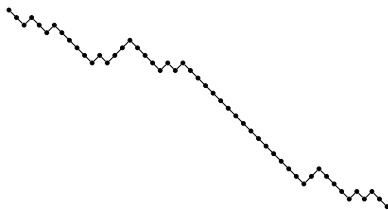


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$$p(\mathbf{y}|\theta) \propto \theta^{12}(1-\theta)^{38}.$$

$$\begin{array}{cc} \uparrow & \uparrow \\ d=12 & \beta=39 \end{array}$$

→ mode → maximum!

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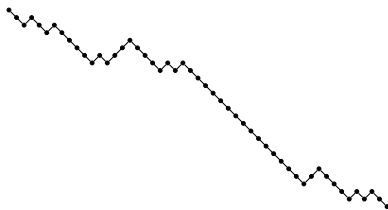


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$$p(\mathbf{y}|\theta) \propto \theta^{12}(1-\theta)^{38}.$$

The corresponding maximum is attained by  $\hat{\theta} = \frac{12}{50} = 0.24$ .

\* up  
tot. observations

## Example: with numbers

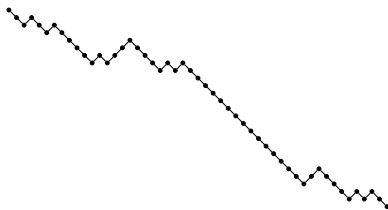


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Conclusion: we expect the next movement to be up with a prob. of 24%.

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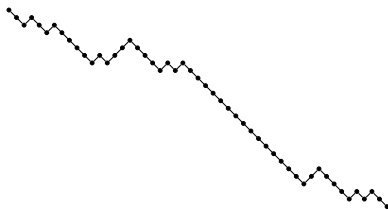


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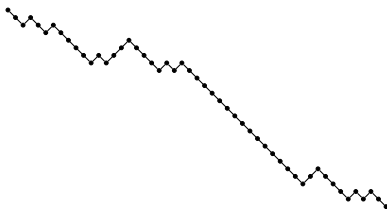


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$\uparrow$   
the estimate

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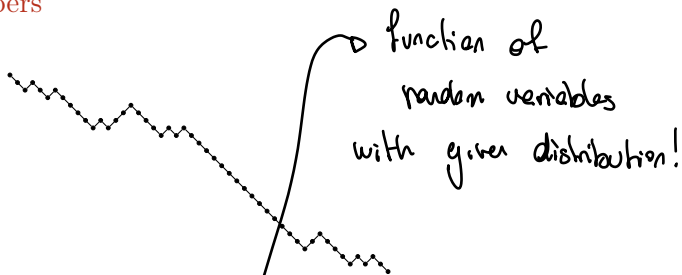


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An **estimator** of  $\theta$  is given by a map  $\mathbf{y} \mapsto \hat{\theta}(\mathbf{y})$  (or just  $\hat{\theta}$ ).

↑  
for every set of observations  
we get a different optimal  
parameter  $\hat{\theta}(\mathbf{y})$ .

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→ Example:  $Y \sim \mathcal{N}(\mu, \sigma^2)$ , then  $\hat{\mu}(\mathbf{y}) = \frac{\sum_{i=1}^n y_i}{n}$ .

↑ input: observations

output: a candidate value for  $\hat{\mu}$ .

→ Example.  $\hat{\Theta}(\mathbf{y}) = \underset{\Theta}{\operatorname{argmax}} \left( \Theta^{\sum y_i} (1-\Theta)^{n - \sum y_i} \right)$

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RV  $\mathbf{Y} = (Y_1, \dots, Y_n)$

$Y_i \sim$  according to  $\theta$

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$$\text{Var}(\hat{\theta}, \theta) = \mathbb{E}[(\hat{\theta}(\mathbf{Y}) - \mathbb{E}[\hat{\theta}(\mathbf{Y})])^2] = \mathbb{E}[\hat{\theta}(\mathbf{Y})^2] - \underbrace{\mathbb{E}[\hat{\theta}(\mathbf{Y})]^2}_{\text{if } \theta \text{ is unbiased, this vanishes}}$$

if  $\theta = a$ , we say that  $\hat{\theta}$  is unbiased

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Example: What about the MLE of  $\mu$  assuming  $\sigma$  given for  $Y \sim \mathcal{N}(\mu, \sigma^2)$ ? Exercise!  
Bias and Variance?

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How to compute it? Equating the derivative to 0.

wrt  $\theta$

if the pdf is differentiable.

Example: with numbers

$$p(y|\theta) = \theta^{\sum y_i} (1-\theta)^{n-\sum y_i}$$

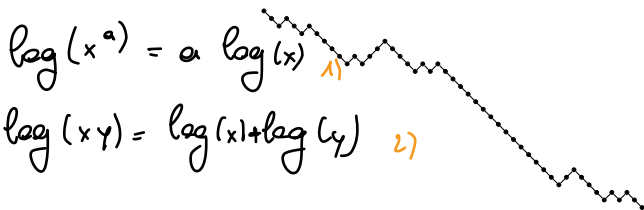


Figure: Price movements

$$\log(p(y|\theta)) \stackrel{2)}{=} \log(\theta^{\sum y_i}) + \log((1-\theta)^{n-\sum y_i})$$

$$\stackrel{1)}{=} \underbrace{(\sum y_i)}_{\text{const}} \log(\theta) + \underbrace{(n - \sum y_i)}_{\text{const}} \log(1-\theta)$$

$$\frac{d}{d\theta} \left( \log(p(y|\theta)) \right) = (\sum y_i) \frac{1}{\theta} + (n - \sum y_i) \frac{-1}{1-\theta} \stackrel{!}{=} 0$$

$$\frac{d}{d\theta} \left( \log(p(y|\theta)) \right) = \left( \sum y_i \right) \frac{1}{\theta} + (n - \sum y_i) \frac{-1}{1-\theta} \stackrel{!}{=} 0$$

$$\cdot \theta(1-\theta) \Big| \Leftrightarrow \left( \sum y_i \right) (1-\theta) - (n - \sum y_i) \theta \stackrel{!}{=} 0$$

$$\Leftrightarrow \sum y_i = \left( \cancel{\sum y_i} + n - \cancel{\sum y_i} \right) \theta$$

$$\Leftrightarrow \frac{\sum y_i}{n} = \theta$$

# up-movements

# total movements

## Example: with numbers

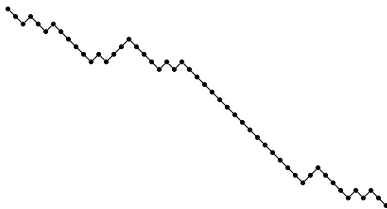
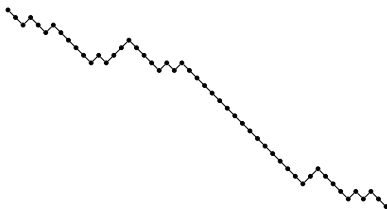


Figure: Price movements

Log likelihood:  $\log(p(\mathbf{y}|\theta)) \propto$

## Example: with numbers



**Figure:** Price movements

Log likelihood:  $\log(p(\mathbf{y}|\theta)) \propto (\sum_{i=1}^n y_i) \log(\theta) + (n - \sum_{i=1}^n y_i) \log((1 - \theta))$ .

## Example: with numbers

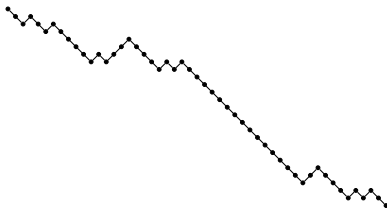


Figure: Price movements

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Differentiating and equation to 0 we get



## Example: with numbers

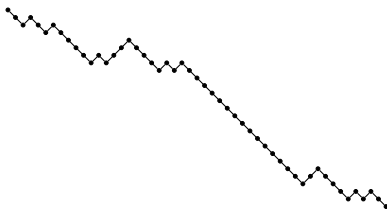


Figure: Price movements

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$$\frac{\sum_{i=1}^n y_i}{\theta} + \frac{n - \sum_{i=1}^n y_i}{\theta - 1} = 0,$$

## Example: with numbers

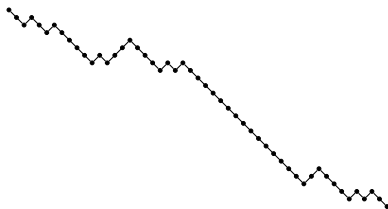


Figure: Price movements

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## Example: with numbers

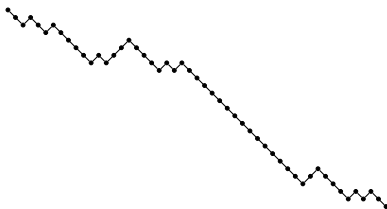


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Its bias and variance are given by

$$\text{bias}(\hat{\theta}, \theta) = \mathbb{E}\left[\frac{\sum_{i=1}^n Y_i}{n}\right] - \theta =$$

$$Y_i \sim \text{Ber}(\theta) \Rightarrow \mathbb{E}[Y_i] = \theta$$

unbias!

$$= \frac{1}{n} \sum_{i=1}^n \mathbb{E}[Y_i] - \theta = \frac{1}{n} \cdot n \cdot \theta - \theta = 0$$

## Example: with numbers

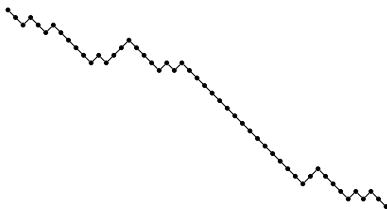


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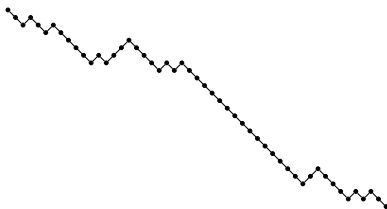


Figure: Price movements

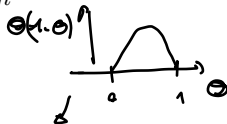
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$$\text{Var}(\hat{\theta}, \theta) = \text{Var}\left(\frac{\sum_{i=1}^n Y_i}{n}\right) = \frac{\sum_{i=1}^n \text{Var}(Y_i)}{n^2} = \frac{\theta(1 - \theta)}{n}.$$



## Combining bias and variance: loss functions

Examples of loss functions:

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**Absolute error:**  $L_1(\hat{\theta}, \theta) = |\hat{\theta} - \theta|$

**Squared error:**  $L_2(\hat{\theta}, \theta) = (\hat{\theta} - \theta)^2$